

Yunshu Chen

Machine Learning Engineer • Medical Imaging, Foundation Models, Agentic AI

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Machine learning engineer and PhD candidate with a software engineering foundation and four years building ML systems that reach production: a distilled segmentation model running on hospital hardware without a GPU, an LLM-orchestrated clinical reporting agent on a locally hosted model, and a cell analysis platform in daily use by an industry partner.

EXPERIENCE

Machine Learning Engineer (Part-time) Jan 2026 – present

PredictX Health × Western Health, Melbourne. Chemotherapy toxicity prediction for colorectal cancer

- Built the clinical data pipeline joining an encrypted registry, CT-derived body composition and chemotherapy administration records into a single source of truth: **302 patients, 57-column schema, 251 strictly matched records**, with five derived clinical outcome variables.
- Established **pytest regression coverage**, an auditable manual-override mechanism, and a data dictionary written for non-technical clinical readers. Feature-selection pipeline (logistic regression, LASSO, Random Forest) identified CT-derived muscle density as the strongest toxicity predictor.

Machine Learning Researcher / Engineer (PhD) Oct 2023 – present

Monash University, with the Victorian Heart Institute, Lead Healthcare and Apollo Medical Imaging

- Agentic clinical reporting.** Architected an LLM-orchestrated reporting system on a **locally hosted Qwen2.5-7B + vLLM** stack (required for patient-data privacy), with a JSON-Schema tool registry and Pydantic state ledger. Every number comes from a deterministic tool; the model plans and narrates but never produces a measurement. Reviewed by clinicians across 35 cases through a review interface I built for them.
- Deployable models.** Knowledge distillation cut a segmentation model **2.6× smaller and 3× faster on CPU** with no measurement degradation, so it runs on hospital hardware without GPUs, on **1/6** of the annotation budget. Uncertainty-guided deferral cut measurement error 30%. Validated against expert measurement at **CCC 0.94 / 0.96** and **84% agreement** on the treatment threshold.
- Production ML at scale.** Trained a cell instance segmentation model on **100K+ images and 8M+ instances** across four imaging modalities, reaching **89% mAP** and outperforming four published models across 12 benchmarks. **Deployed as a production web application**, with LoRA fine-tuning so users onboard a new cell type from 5 to 10 samples.
- Data infrastructure.** Built a **49.6M-frame** DICOM to tensor pre-training pipeline with md5-versioned data lineage, multi-GPU training and an experiment dashboard, plus an 8-GPU crash-safe evaluation framework; data audits caught duplication and corruption in widely used public datasets.

Speech Recognition Algorithm Engineer, Intern (Full-time) Jul 2022 – Dec 2022

NIO, Intelligent Cockpit & Software Development, Shanghai, China

- Improved multilingual ASR accuracy by **5%** across Chinese and European languages; built automated diagnostics that resolved **60+ production recognition failures**; **integrated models into production vehicle systems** alongside product, UX and NLU teams.

SELECTED PROJECTS

- Genius Labbook** (open source, github.com/CHEN-Yunshu/Genius_Labbook). Python CLI recording experiments, results, figures and decisions as Markdown + YAML with automatic git commits; packaged with pyproject/entry_points, full-text search and figure archiving. Has driven all of my research since March 2026: **135 entries across 6 projects**.
- End-to-Same-End Encryption mobile application** (honours thesis, 2023). Cross-platform iOS/Android secure storage with client-side encryption before upload, so the server never sees plaintext. Full stack: **React Native, Spring Boot, Node.js, AWS S3, Jenkins CI/CD**.

EDUCATION

PhD, Data Science & Artificial Intelligence, Monash University. Multi-modal medical imaging, VLMs, segmentation Oct 2023 – Apr 2027 (expected)

Bachelor of Advanced Computing, Honours II-1, The University of Sydney. Major: Software Development & Computational Data Science Feb 2019 – Jun 2023

TECHNICAL SKILLS

Machine learning	PyTorch · scikit-learn · SAM / SAM2 · DINOv3 · ViT · nnU-Net · MONAI · LoRA · distillation
LLM & agents	ReAct orchestration · vLLM · Qwen2.5 · JSON-Schema tool contracts · local/private deployment
Data & MLOps	large-scale DICOM pipelines · data lineage · multi-GPU training · Docker · Kubernetes · LASSO / Random Forest · A/B testing
Cloud & DevOps	AWS (S3, SageMaker) · Azure · Jenkins CI/CD · Git · Linux · pandas · pytest
Software	Python (advanced) · Java · C++ · SQL · R · React Native · Spring Boot · Node.js · PyQt
Languages	English (professional) · Mandarin Chinese (native)

PUBLICATIONS & AWARDS

Three first-author papers under review in medical imaging and computer vision; one published as co-author. Our team placed **2nd** and **3rd** in the two tracks of the Intelligent Cockpit Speech Recognition Challenge, Singapore 2022.